SG No. 23 D_2^8 $I222$ [orthorhombic]

* plus set: $+[0, 0, 0]$, $+\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$
* Wyckoff site: `2a`, site symmetry: `222`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,-3,-4]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, 0, 0]$	`[1,-2,3,-4]`

Table 3: Wyckoff bond: `2c@2a`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,-2,-3,4]`

Table 4: Wyckoff bond: `4d@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2]`
2	$[-X, Y, 0]$	$[0, 0, 0]$	`[3,-4]`

Table 5: Wyckoff bond: `4e@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-3]`
2	$[-X, 0, Z]$	$[0, 0, 0]$	`[2,-4]`

Table 6: Wyckoff bond: `4f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$[0, 0, 0]$	`[2,-3]`

Table 7: Wyckoff bond: `8g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[0, 0, 0]$	`[3]`

continued ...

Table 7

No.	vector	center	mapping
4	$[X, -Y, -Z]$	[0, 0, 0]	[4]

* Wyckoff site: 2b, site symmetry: 222

Table 8: Wyckoff bond: 2a@2b, bond symmetry: 222

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,2,-3,-4]

Table 9: Wyckoff bond: 2b@2b, bond symmetry: 222

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,-2,3,-4]

Table 10: Wyckoff bond: 2c@2b, bond symmetry: 222

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,-2,-3,4]

Table 11: Wyckoff bond: 4d@2b, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,-2]
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	[3,-4]

Table 12: Wyckoff bond: 4e@2b, bond symmetry: .2.

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,-3]
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[2,-4]

Table 13: Wyckoff bond: 4f@2b, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[1,-4]
2	$[0, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	[2,-3]

Table 14: Wyckoff bond: `8g@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4]`

* Wyckoff site: `2c`, site symmetry: `222`

Table 15: Wyckoff bond: `2a@2c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 16: Wyckoff bond: `2b@2c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,3,-4]`

Table 17: Wyckoff bond: `2c@2c`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 18: Wyckoff bond: `4d@2c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-4]`

Table 19: Wyckoff bond: `4e@2c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-4]`

Table 20: Wyckoff bond: `4f@2c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-3]`

Table 21: Wyckoff bond: `8g@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4]`

* Wyckoff site: `2d`, site symmetry: `222`

Table 22: Wyckoff bond: `2a@2d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,-3,-4]`

Table 23: Wyckoff bond: `2b@2d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,3,-4]`

Table 24: Wyckoff bond: `2c@2d`, bond symmetry: `222`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,-3,4]`

Table 25: Wyckoff bond: `4d@2d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[3,-4]`

Table 26: Wyckoff bond: `4e@2d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-4]`

Table 27: Wyckoff bond: `4f@2d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-3]`

Table 28: Wyckoff bond: `8g@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4]`

* Wyckoff site: `4e`, site symmetry: `2..`

Table 29: Wyckoff bond: `4a@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$[-x, 0, 0]$	`[2,-3]`

Table 30: Wyckoff bond: `4b@4e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3]`

Table 31: Wyckoff bond: `8c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`

* Wyckoff site: `4f`, site symmetry: `2..`

Table 32: Wyckoff bond: `4a@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,-3]`

Table 33: Wyckoff bond: `4b@4f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2,3]`

Table 34: Wyckoff bond: `8c@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	`[4]`

* Wyckoff site: `4g`, site symmetry: `.2.`

Table 35: Wyckoff bond: `4a@4g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, y, 0]$	`[1,-3]`
2	$[-X, 0, Z]$	$[0, -y, 0]$	`[2,-4]`

Table 36: Wyckoff bond: `4b@4g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, y, 0]$	`[1,3]`
2	$[0, -Y, 0]$	$[0, -y, 0]$	`[2,4]`

Table 37: Wyckoff bond: `8c@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, 0]$	`[4]`

* Wyckoff site: `4h`, site symmetry: `.2.`

Table 38: Wyckoff bond: `4a@4h`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,-4]`

Table 39: Wyckoff bond: `4b@4h`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2,4]`

Table 40: Wyckoff bond: `8c@4h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, 0\right]$	`[4]`

* Wyckoff site: `4i`, site symmetry: `..2`

Table 41: Wyckoff bond: `4a@4i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-X, Y, 0]$	$[0, 0, -z]$	`[3,-4]`

Table 42: Wyckoff bond: `4b@4i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2]`
2	$[0, 0, -Z]$	$[0, 0, -z]$	`[3,4]`

Table 43: Wyckoff bond: `8c@4i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-X, Y, -Z]$	$[0, 0, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[0, 0, -z]$	`[4]`

* Wyckoff site: `4j`, site symmetry: `..2`

Table 44: Wyckoff bond: `4a@4j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,-4]`

Table 45: Wyckoff bond: `4b@4j`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[3,4]`

Table 46: Wyckoff bond: `8c@4j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, -z\right]$	`[4]`

* Wyckoff site: `8k`, site symmetry: `1`

Table 47: Wyckoff bond: `8a@8k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -y, -z]$	`[4]`